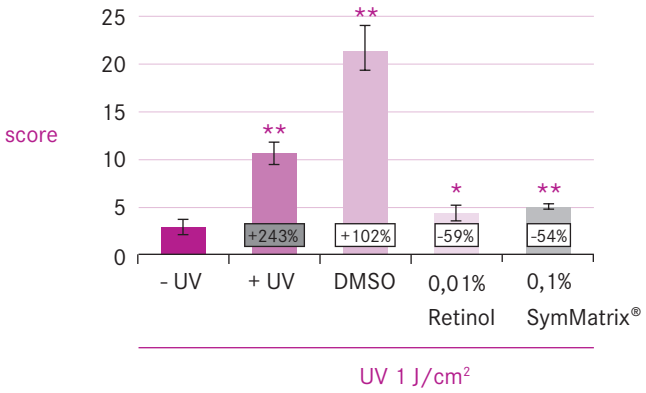
Elastin quantification



☐ Δ % vs Not treated +UV
■ Δ % vs Not treated -UV

Significance vs UV treated
* p< 0.05, ** p<0,01

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symrise
always inspiring more...

ACTIVE INGREDIENTS

SymMatrix®
112222

SymMatrix® No. 112222
Plant-Derived MMP Inhibitor
Ecocert Approved

BENEFITS

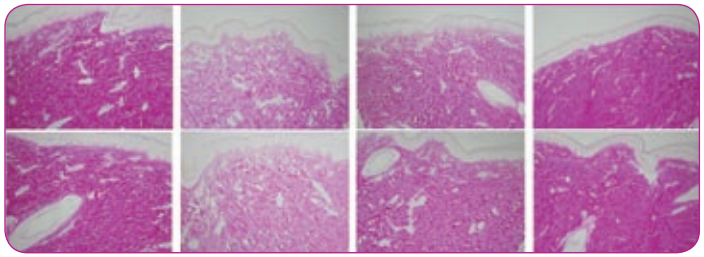
- High efficacy even at low usage concentrations:
  95 % MMP-1 inhibition at 0.01 %
  85 % MMP-9 inhibition at 0.1 %
- Protects against UVA-induced MMP-1 expression
- Antioxydant properties
- Collagen stimulating activity
- Modulates the UV-induced upregulation of elastin
- Like Retinol, SymMatrix® stimulates the dermis metabolism by inhibition of MMPs and elastase, helps to maintain the extra cellular matrix proteins balance
- SymMatrix® provides a multilevel protection against premature skin aging
- Natural raw material approved by Ecocert SAS F 32600 according to the Ecocert standards for Natural and Organic Cosmetics

PRODUCT DESCRIPTION

- Preservative free powder, based on an aqueous-ethanolic extraction of blackberry (Rubus fruticosus) leaves, followed by spray-drying on maltodextrin
- Usage concentration: 0.1 to 1.0%
  - SymMatrix® contains Polyphenols: formulate at pH <5
- Beige to brown
- Water-soluble
- Patent pending
- INCI: Maltodextrin, Rubus Fruticosus (Blackberry) Leaf Extract

COLLAGEN EVALUATION
ex vivo skin pig study

Miller's stain of collagen. Representative picture of treated samples.

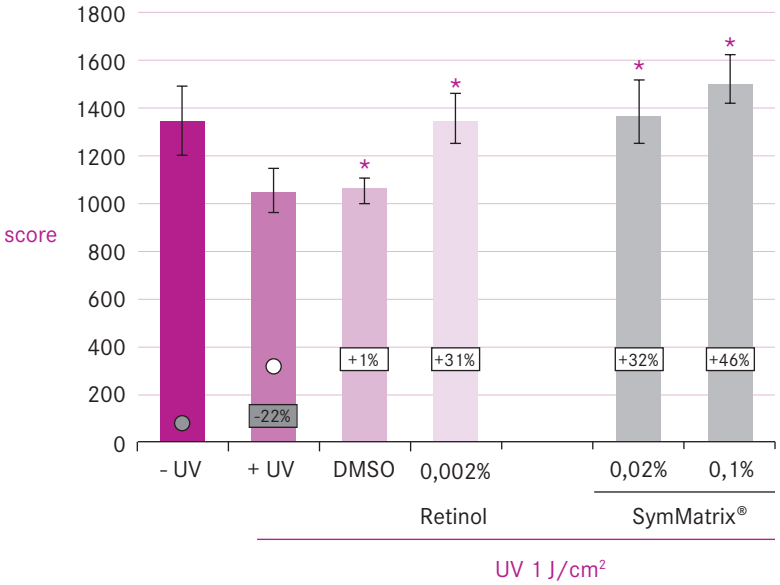


- UV | + UV | SymMatrix® 0,02% | SymMatrix® 0,1%

RESULTS

- As expected, the samples irradiated with UV and then treated with medium (+ UV) or DMSO showed clear collagen degradation
- Both, retinol and SymMatrix® treatment led to significantly higher collagen levels than vehicle (DMSO) treatment.
- 0,002% retinol and 0,02% SymMatrix® showed comparable activity

ELASTIN EVALUATION
Collagen quantification



□ Δ % vs Not treated +UV
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*Significance vs UV treated p < 0,05